Experiment No. 2	
Edge Computing Lab	
Name of Student:	
Class :	Date of Performance:
Roll No :	Date of Checking :
	Sign :

Title: IoT-Based Temperature and Humidity Monitoring using Raspberry Pi 5 and Blynk Cloud

Objective

To interface a DHT11 sensor with a **Raspberry Pi 5** and transmit real-time environmental data to the **Blynk IoT Cloud** for remote visualization on a web and mobile dashboard.

Background

In advanced IoT ecosystems, data processed at the edge (Raspberry Pi) is often pushed to the cloud for long-term storage and remote monitoring.

- **Blynk IoT:** A platform that allows developers to create custom dashboards using "Virtual Pins" to represent physical sensor data.
- **RP1 Architecture:** The Pi 5's dedicated I/O controller requires libgpiod to manage sensor timing, which is critical for the DHT11's proprietary one-wire protocol.

Materials Required

- Raspberry Pi 5 (with Internet Connection)
- DHT11 Temperature and Humidity Sensor
- Jumper Wires (Female-to-Female)
- 4.7kΩ - 10kΩ Resistor (if using a raw sensor)
- Blynk IoT Account (Cloud Console)

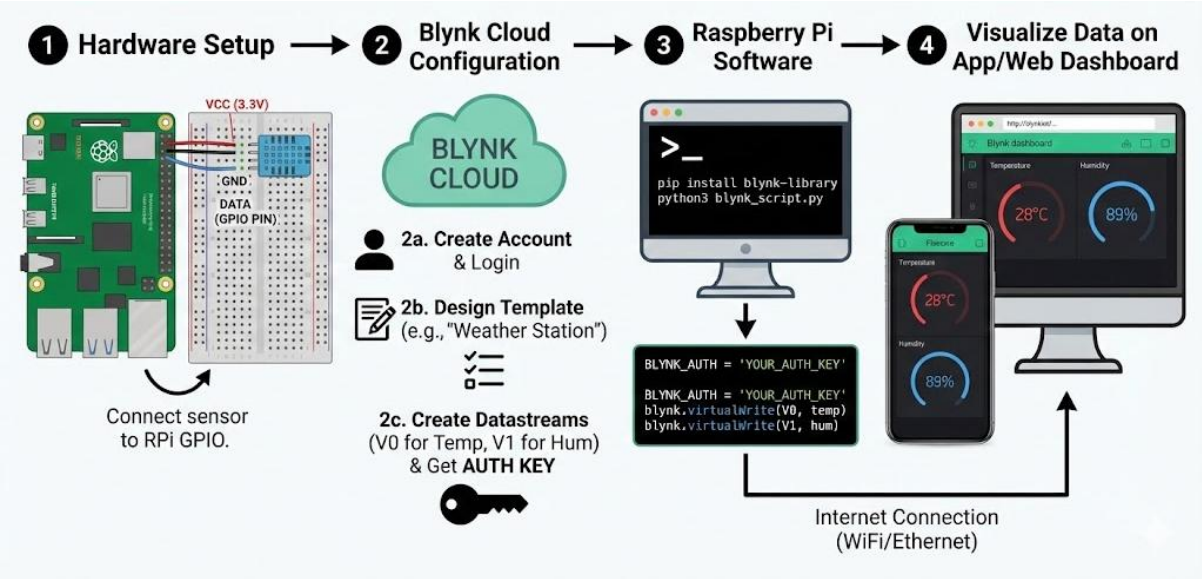
Circuit Diagram & Wiring

CRITICAL: Ensure the sensor is powered via the **3.3V** pin. The Raspberry Pi 5 GPIO pins are **not 5V tolerant**; using the 5V rail for data signaling will cause permanent damage to the RP1 chip.

Sensor Pin	Function	Raspberry Pi 5 Pin	Physical Pin #
VCC (+)	Power	3.3V	Pin 1
GND (-)	Ground	Ground	Pin 6
DATA (Out)	Signal	GPIO 4	Pin 7

Block Diagram

The system follows a **Cloud-Edge Architecture** where the Pi 5 acts as the edge gateway.



Software Setup Procedure

Step 1: Environment Setup

On the Pi 5 (OS Bookworm), we must use a virtual environment to manage dependencies reliably.

```bash
 1. Create a project folder
mkdir blynk_dht11 && cd blynk_dht11

2. Create and activate virtual environment
python3 -m venv --system-site-packages venv
source venv/bin/activate

3. Install necessary libraries
pip install blynk-library-python adafruit-circuitpython-dht
```

Implementation (Python Script)

Create blynk_dht11.py and insert the code below. Replace the BLYNK_AUTH with your specific token.

Python

```python
import BlynkLib
import board
import adafruit_dht
import time

# --- BLYNK CONFIGURATION ---
# Replace with your actual Auth Token from Blynk Device Info
BLYNK_AUTH = '0aQ7ypo4j-OCPek4-fwQ7Phq1Wxxxxxx'

# Force connection to the new Blynk IoT Cloud
blynk = BlynkLib.Blynk(BLYNK_AUTH, server='blynk.cloud', port=80)

# --- SENSOR CONFIGURATION ---
# DHT11 connected to GPIO 4 (Physical Pin 7)
sensor = adafruit_dht.DHT11(board.D4)

print("Starting Blynk DHT11 Monitoring on Pi 5...")

def send_sensor_data():
    try:
        # Read values from sensor
        temp = sensor.temperature
        humi = sensor.humidity

        if humi is not None and temp is not None:
            # Print to local console for debugging
            print(f"Blynk Update -> Temp: {temp:.1f}°C | Hum: {humi:.1f}%")

            # Send to Blynk Virtual Pins (V0 for Temp, V1 for Hum)
            blynk.virtual_write(0, temp)
            blynk.virtual_write(1, humi)
        else:
            print("Sensor returned null reading...")

    except RuntimeError as error:
        # DHT timing errors are common on Pi 5; we catch and continue
        pass
    except Exception as error:
        sensor.exit()
        raise error

# Main Loop
last_send_time = 0
while True:
    blynk.run()

    # Send data every 5 seconds to avoid flooding the Blynk server
    current_time = time.time()
    if current_time - last_send_time > 5.0:
        send_sensor_data()
        last_send_time = current_time
```

Execution

1. **Activate VENV:** source venv/bin/activate
2. **Run Script:** python3 blynk_dht11.py
3. **View Dashboard:** Open your Blynk Mobile App or Web Console to see the live gauges updating.

Theory & Review Questions

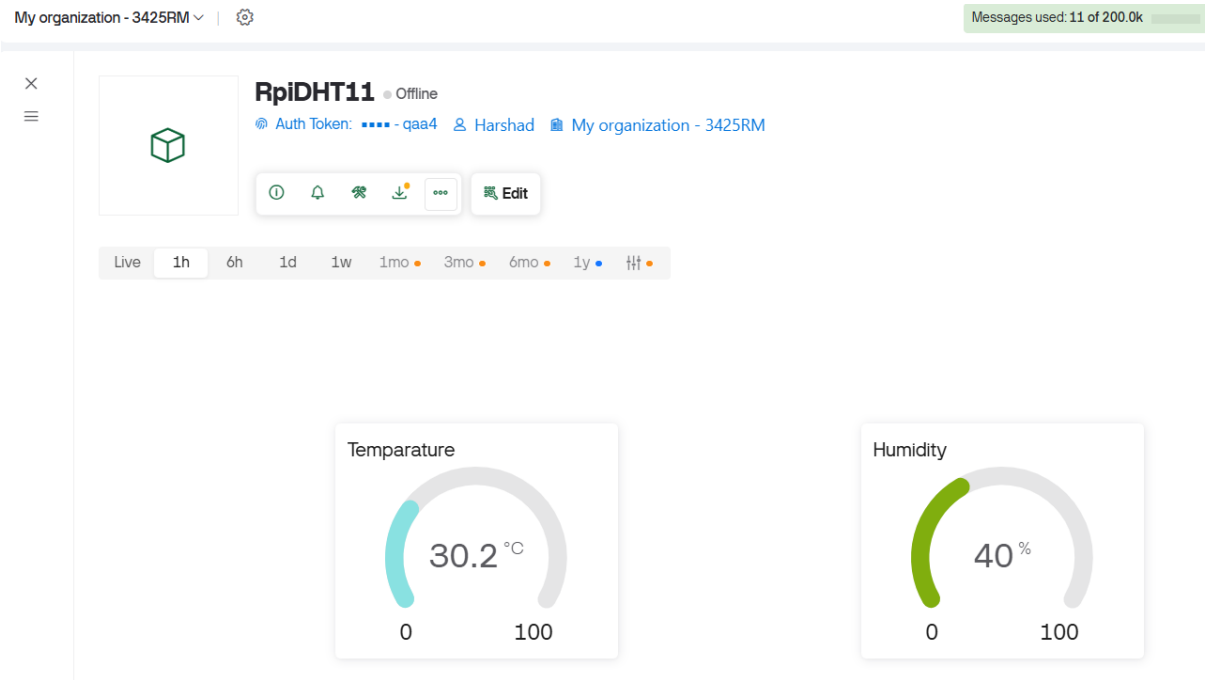
Q1. Explain why it is necessary to specify the server='blynk.cloud' and port=80 parameters in the code for the latest Blynk IoT platform.

Q2. How does the blynk.run() function operate within the main loop, and why should we avoid using time.sleep() inside the same loop?

Q3. Describe the difference between a Physical Pin and a **Virtual Pin (V0, V1)** in the context of Blynk IoT.

Expected Output

- **Console:** Displays "Blynk Update -> Temp: XX.X°C | Hum: XX.X%" every 5 seconds.
- **Blynk Dashboard:** Gauges or charts showing real-time environmental fluctuations.



Conclusion